

Regularizing Dispersion Estimates in Negative Binomial Models for Improved scRNA-seq Analysis

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Abstract

Single-cell RNA sequencing count data is overdispersed and difficult to model. Poisson models fail to account for overdispersion, and no clear consensus exists on parameterization methods for the Negative Binomial. Four θ regularization strategies accounting for mean gene expression and overdispersion were evaluated on PBMC data: global, gene-wise, binned, and LOESS. The performance of each strategy was evaluated using chi-squared goodness-of-fit, variance stabilization, and PCA. Binned and LOESS regularization strategies reduced poorly fit genes from 26.5% to 11%. Gene-wise estimates of θ were prone to overfitting, and global estimates failed to accurately capture dispersion in the data. Mean-expression-based regularization was found to improve both statistical fit and biological interpretability.

1 Introduction

Single-cell RNA sequencing (scRNA-seq) has become an invaluable method in bioinformatics, enabling high-resolution profiling of gene expression at the cellular level [1]. However, the resulting data is computationally challenging to analyze, due to sparsity, technical noise, and variability in sequencing depth [2]. In particular, scRNA-seq is often applied to biologically heterogeneous systems, where observed variation arises from both true biological differences (e.g., cell type and state) and technical or stochastic effects. Separating these sources of variation during preprocessing remains a central challenge [3]. To address this, statistical modeling approaches have been developed to describe the distribution of scRNA-seq count data. These methods aim to jointly capture biologically meaningful variation through a latent representation while accounting for technical and stochastic noise with an appropriate error model. There is currently no consensus on the most appropriate error distribution or the estimation of its parameters [3]. In particular, accurately modeling overdispersion remains a challenge in scRNA-seq analysis.

One common approach is to model scRNA-seq counts using a Poisson distribution. The Poisson distribution assumes the variance of the data equals the mean. However, this assumption is often violated in practice, due to cell heterogeneity and variability in gene expression, resulting in overdispersed data [3]. As a result, the Poisson model frequently underestimates

variability and provides a poor fit for many genes. Another approach is the Negative Binomial (NB) model, which introduces an inverse overdispersion parameter, θ , removing the constraint that variance equals the mean. This method generally provides a better fit for scRNA-seq data, but its parameterization remains unclear. Prior work has shown that using a fixed global value of θ does not generalize well across datasets, and estimating θ independently for each gene is unstable, especially for lowly expressed genes.

This suggests that dispersion should be modelled based on gene-specific properties, such as average gene expression level. This project seeks to evaluate statistical error models for scRNA-seq data, focusing on how different parameterizations of the Negative Binomial dispersion parameter (θ) affect model fit and downstream analysis.

2 Methods

All analysis was implemented in Python; code is available in the Section 5.

2.1 Data sources and preprocessing

A peripheral blood mononuclear cell (PBMC) dataset obtained as a preprocessed count matrix from the 10x Genomics Chromium platform was used for analysis [4]. Genes expressed in fewer than 5 cells, and cells with fewer than 200 genes detected were filtered out of the dataset. Quality control metrics (counts per cell, genes per cell, and cells per gene) were inspected to guide filtering thresholds (see Appendix A.1). Each cell was normalized to a total of 10 000 counts to account for differences in sequencing depth. Raw counts were stored separately for downstream goodness-of-fit evaluations.

2.2 Dispersion Estimation

Dispersion parameters were estimated under the Negative Binomial model, which assumes a mean-variance relationship of the form:

$$\text{Var}(X) = \mu + \frac{\theta}{\mu^2} \quad (1)$$

Analysis was restricted to over-dispersed genes ($\text{Var} > \mu$) to ensure valid estimates. A Poisson model was used as a baseline, assuming variance was equal to the mean. Four approaches to dispersion estimation were analyzed:

- **Gene-wise θ (method-of-moments):** Dispersion was independently for each gene using (1)
- **Global θ :** A single dispersion parameter was computed as the median of the computed gene-wise θ estimates and applied across all genes.
- **Binned θ :** Genes were grouped into 50 equal-frequency bins based on log mean expression. The median θ was assigned as the shared dispersion value for each bin.
- **LOESS θ :** A smooth curve fit in log-log space relating mean expression to θ was estimated using LOESS regression. The computed function was then interpolated back to each gene.

2.3 Model Evaluation

Model performance was evaluated using two criteria: (1) Variance stabilization was assessed using the variance of Pearson residuals across genes. An ideal model produces an approximately constant variance. (2) Principal component analysis (PCA) was used on residuals to assess the separation of cell populations. Improved parameter estimation is expected to produce clearer cluster separation.

3 Results

3.1 Mean-variance relationship

The relationship between gene-wise mean and variance was examined to assess the applicability of the Poisson model. As shown in Figure 1, variance increases with mean expression in a non-linear fashion. This relationship violates the Poisson assumption of equal mean and variance, indicating overdispersion in the data. This overdispersion motivates the use of a Negative Binomial model.

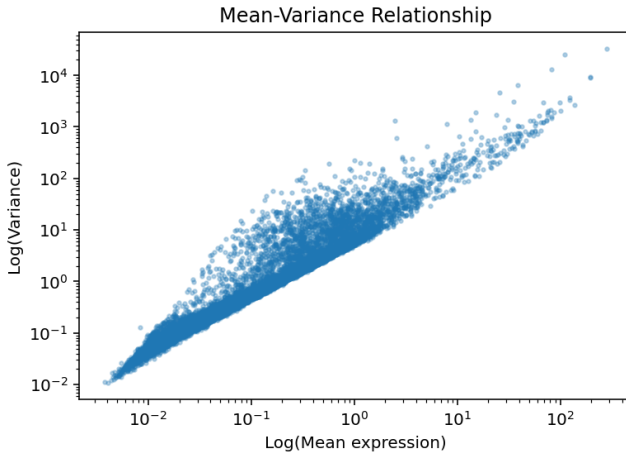


Figure 1: Genes with increased mean expression tend to have increased variance in counts.

3.2 Dispersion estimation behavior

Gene-wise dispersion estimates displayed significant variability, particularly for lowly expressed genes (Figure 2). Furthermore, dispersion was found to strongly depend on mean expression, with overdispersion decreasing as mean expression increases. As shown in Figure 2, the global dispersion model fails to capture the observed trend. Binning reduced noise, but introduced discontinuities. In contrast, LOESS-based smoothing resulted in a continuous and biologically sound relationship between dispersion and mean gene expression.

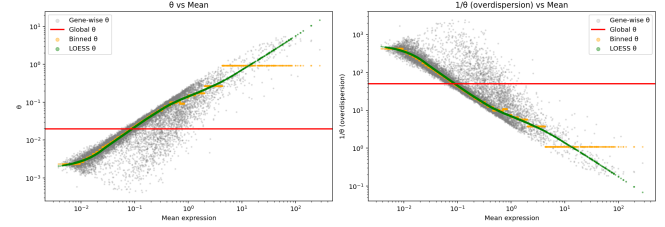


Figure 2: Lowly expressed genes are associated with increased overdispersion. Gene-wise θ estimates are noisy, especially for lowly expressed genes. A global θ parameter fails to reflect the empirical trend of the data. Binned and LOESS θ parameters better capture this trend, but binning causes discontinuities in estimates.

3.3 Goodness of Fit

A chi-squared goodness-of-fit test (Figure 3) was performed on all models to evaluate the accuracy of each model's assumed variance formula. An ideal model has $\chi^2/df \approx 1$, indicating its variance formula matches the spread of counts across cells per gene.

The Poisson model's distribution is spread far to the right, resulting in 100% of genes having $\chi^2/df > 1$, indicating severe underfitting. The global θ parameter performed better than Poisson, but peaks to the left of 1 with a heavy right tail, showing signs of overcorrection for many genes. Gene-wise, Binned, and LOESS all peak very close to 1; gene-wise parametrization is artificially perfect, indicated by a standard deviation of zero (Table 1). Binned and LOESS are virtually indistinguishable quantitatively, both reducing poorly fit genes from 26.5% (Global) to 11% (Table 1). This similarity suggests that smooth continuous regularization doesn't meaningfully outperform a simpler binning approach.

Table 1: Summary statistics of Pearson χ^2 test per model

Model	Med	Mean	Std	P90	% > 2
Poisson	5.50	9.77	17.93	16.05	100
Global θ	1.00	1.52	2.03	3.08	26.6
Gene-wise θ	1.00	1.00	0.00	1.00	0
Binned θ	1.00	1.60	2.87	2.19	11.0
LOESS θ	0.98	1.59	2.83	2.20	11.0

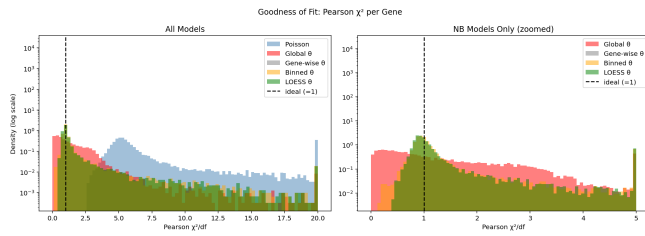


Figure 3: Pearson χ^2 for various θ parametrization methods. Ideal distributions will peak at one, indicating poor fit of Poisson and global θ models.

3.4 Count Distribution

For MALAT1, a highly expressed gene, the Poisson model produces a narrow symmetric curve centred about the mean that fails to capture the heavy right tail of the observed count distribution (Figure 12). The LOESS model, however, produces a right-skewed NB curve that resembles the histogram shape. For RPS23, a lowly expressed gene, a similar pattern holds, but the failure modes differ slightly. The Poisson curve again misses the tail, while LOESS captures the overall shape reasonably well. However, neither model perfectly captures the spike of near-zero counts, which likely reflects the high dropout rate typical of lowly expressed genes in scRNA-seq data.

The Global θ failure is visually evident in both genes with the fitted curve collapsing to near zero across almost the entire count range, confirming that a single dispersion parameter shared across all genes is insufficient.

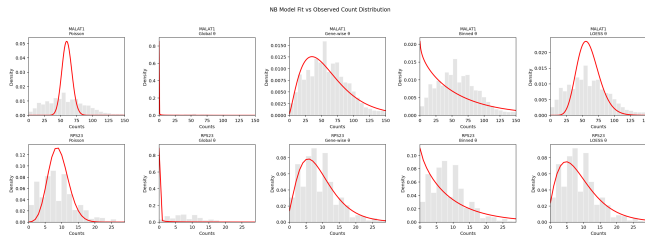


Figure 4: Fitted NB and Poisson distributions overlaid on observed count histograms for MALAT1 (highly expressed) and RPS23 (lowly expressed).

3.5 Model Evaluation

The variance of the Pearson residuals from each distribution was used to assess model fit. Residual variance closer to 1 indicates the θ parametrization of the model sufficiently captures how much genes vary across cells. As seen in Figure 5, the Poisson model was not able to accurately estimate the variance in the data. Models that incorporated gene-specific θ parameterization produced residuals with variance closer to one, indicating improved modelling of gene-specific variability in the data.

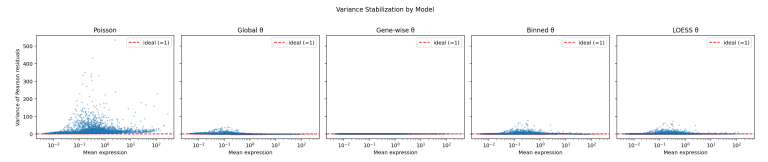


Figure 5: Variance of Pearson residuals per gene across mean expression levels. Points closer to the red dashed line ($=1$) indicate better variance stabilization.

3.6 PCA Analysis

As the variance stabilization plots for the different methods of θ parametrization were similar, PCA analysis (Figure 6) was performed on model-derived residuals to better visualize the impact of parametrization on downstream analysis. The global dispersion model produced poorly separated clusters, indicating poor modelling of variability. The Poisson PCA plot spans a larger axis than the gene-specific θ parameterization methods, indicating poor outlier suppression. Gene-wise estimates partially recover structure but remain noisy.

In contrast, binned and LOESS-based dispersion models yield clearer separation of cell populations. This suggests that modelling dispersion as a function of mean expression improves both variance stabilization and preservation of biologically meaningful structure.

4 Discussion

4.1 Interpretation

The obtained results indicate that binned and LOESS regularization methods performed better than other methods, and also obtained similar results. This suggests that biological variation is better captured using when accounting for the relationship between mean gene expression and dispersion. The smoothness of the method does not seem to be a crucial factor; thus, for practical approaches, the binning method may be simpler.

Gene-wise estimation of dispersion parameters produced artificially optimal goodness-of-fit metrics, with $\chi^2/df \approx 1.000$ and negligible variance across genes. While this may initially suggest superior model fit, it reflects overfitting rather than true model accuracy. Gene-wise models adapt to the noise within each gene, eliminating biologically meaningful variation and generalization. These results are consistent with findings by Choudry and Satija and emphasize the need for regularization to obtain meaningful dispersion estimates. These findings are also consistent with prior work demonstrating the importance of regularized dispersion estimation in scRNA-seq analysis, as seen in methods such as SCTransform.

4.2 Limitations

Dispersion parameters were estimated using a method-of-moments approach, which is known to be less stable than max-

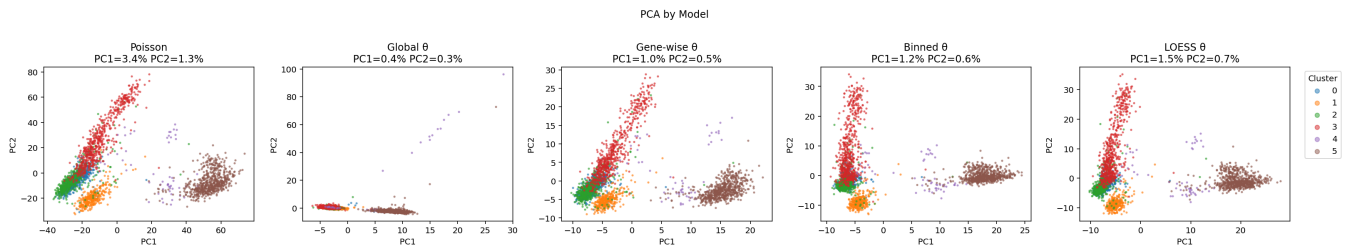


Figure 6: PCA of residuals under different dispersion models. Regularized approaches (binned and LOESS) produce clearer separation of cell populations compared to Poisson, global, and gene-wise models.

imum likelihood estimation, particularly for lowly expressed genes. This likely contributed to the high variability observed in gene-wise dispersion estimates.

Secondly, the Negative Binomial model does not explicitly account for zero inflation, which is common in scRNA-seq data due to technical dropout. This limitation was evident in the count distributions of certain genes (e.g., RPS23), where observed zero frequencies exceeded model predictions.

Third, downstream evaluation using PCA relied on clustering structure derived from the same dataset. While PCA provides a useful qualitative assessment of biological structure, independent validation on external datasets would strengthen the conclusions.

5 Conclusions

The Poisson model does not accurately model the distribution of scRNA-seq count data due to overdispersion. While a global dispersion parameter improves fit relative to Poisson, it fails to capture the relationship between dispersion and mean expression. In contrast, simple regularization strategies, including binning and LOESS, substantially improve both model fit and downstream clustering while avoiding the overfitting in gene-wise estimation. The similar results from these approaches suggest that capturing the mean-dispersion relationship is crucial. Future work may consider maximum likelihood estimation or account for zero-inflation.

Data Availability

The code used for data processing, model implementation, and analysis is publicly available at: <https://github.com/joannwokeforo/Regularizing-Dispersion-Estimates-in-Negative-Binomial-Models.git>

References

- [1] H. M. A. M. Nesari and S. Ghorbian, “Advances and challenges in single-cell rna sequencing data analysis: a comprehensive review,” *Brief Bioinform*, vol. 27, no. 1, pp. 1–20, Jan. 2026.

- [2] A. S. S. D. C. A. Vallejos, D. Risso and J. C. Marioni, “Normalizing single-cell rna sequencing data: challenges and opportunities,” *Nat Methods*, vol. 14, no. 6, pp. 565–571, Jun. 2017.
- [3] S. Choudhary and R. Satija, “Comparison and evaluation of statistical error models for scRNA-seq,” *Genome Biol*, vol. 23, no. 27, pp. 1–20, Jan. 2022.
- [4] (2016, May) 3k pbmcs from a healthy donor. Online. 10x Genomics. [Online]. Available: <https://www.10xgenomics.com/datasets/3-k-pbmcs-from-a-healthy-donor-1-standard-1-1-0>

A Appendix

A.1 Preprocessing Graphs

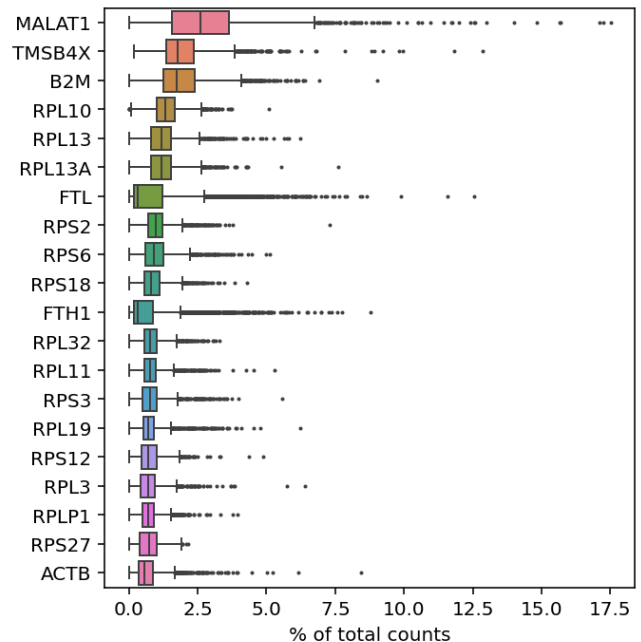


Figure 7: Top 20 Highly Expressed Genes

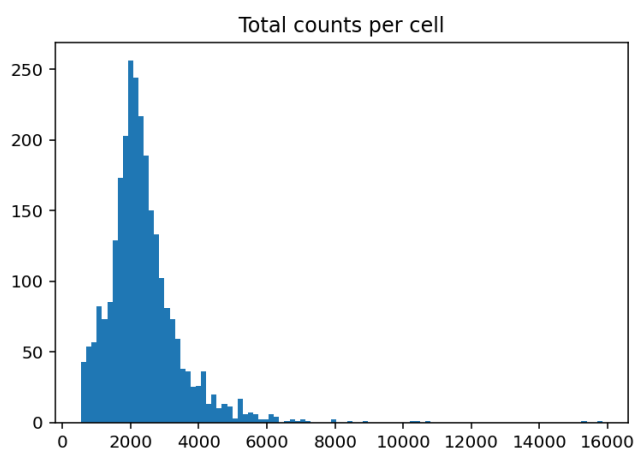


Figure 8: Total counts per cell

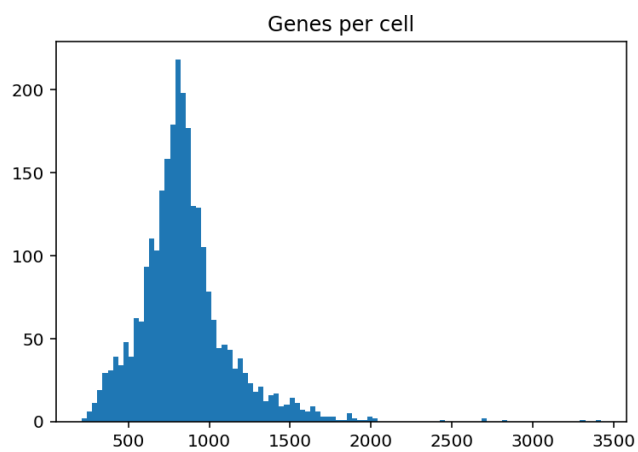


Figure 11: Initial genes per cell counts.

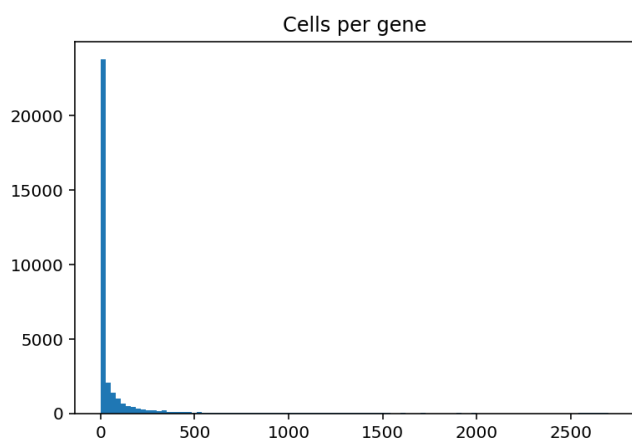


Figure 9: Initial cells per gene counts.

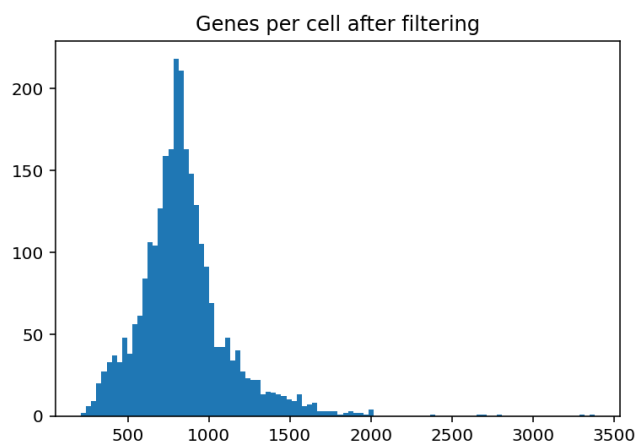


Figure 12: Filtered genes per cell counts

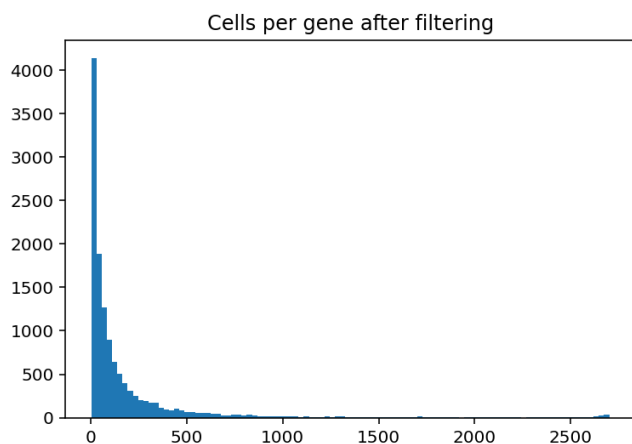


Figure 10: Filtered cells per gene counts.